

8. To analyse the stick-free dynamic response of the aircraft to a perturbation, the elevator deflection δ_E , must be added as an additional degree of freedom to our system of longitudinal equations of motion. Starting from the longitudinal equations of motion

$$m\dot{U} = X^a - mg \sin \Theta \quad (1)$$

$$m(\dot{W} - Uq) = Z^a + mg \cos \Theta \quad (2)$$

$$I_{yy}\dot{q} = M^a \quad (3)$$

and the elevator hinge moment equation

$$I_E \ddot{\delta}_E = H_E = \frac{1}{2} \rho V_\infty^2 S_E c_E (b_0 + b_H \alpha_H + b_E \delta_E + b_T \delta_T), \quad (4)$$

use small-perturbation theory to show that the state-space representation of the aircraft's longitudinal motion starting from an equilibrium, level flight condition, is

$$\begin{matrix} (5) \\ (6) \end{matrix} \begin{bmatrix} m & -X_{\dot{W}} & 0 & 0 & 0 & 0 \\ 0 & m - Z_{\dot{W}} & 0 & 0 & 0 & 0 \\ 0 & -M_{\dot{W}} & I_{yy} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -H_{\dot{W}} & 0 & 0 & I_E & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{U} \\ \dot{W} \\ \dot{q} \\ \dot{\Theta} \\ \ddot{\delta}_E \\ \dot{\delta}_E \end{bmatrix} = \begin{bmatrix} X_U & X_W & X_q & -mg & 0 & X_{\delta_E} \\ Z_U & Z_W & Z_q + mU_e & 0 & 0 & Z_{\delta_E} \\ M_U & M_W & M_q & 0 & 0 & M_{\delta_E} \\ 0 & 0 & 1 & 0 & 0 & 0 \\ H_U & H_W & H_q & 0 & 0 & H_{\delta_E} \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} U \\ W \\ q \\ \Theta \\ \dot{\delta}_E \\ \delta_E \end{bmatrix}.$$

You may assume that the effect of the elevator's rate of rotation and rotational acceleration on the aircraft's aerodynamics is negligible.

For small perturbation theory,

$$\mathbf{x} = [U \ W \ q \ \Theta \ \dot{\delta}_E \ \delta_E]$$

Apply a small perturbation $\mathbf{x}'$ at equilibrium point

$$\mathbf{x}' = [U' \ W' \ q' \ \Theta' \ \dot{\delta}_E' \ \delta_E']$$

① *Therefore*

$$m(\dot{U}_0 + \dot{U}') = X_0^a + U X_U^a + W X_W^a + \dot{W} X_{\dot{W}}^a + q X_q^a - mg \sin(\Theta_0 + \Theta')$$

$$\text{where } mg \sin(\Theta_0 + \Theta') \approx mg(\Theta_0 + \Theta')$$

$$\text{and } m\dot{U}_0 = X_0^a - mg\Theta_0 = 0$$

② also,

$$m(\dot{W}_0 + \dot{W}') = m(U_0 + U')(q_0 + q') \\ + Z_0^a + UZ_0^a + WZ_W^a + \dot{W}Z_W^a + qZ_q^a + mg \cos(\theta_0 + \theta')$$

where $mg \cos(\theta_0 + \theta') \approx mg$

and $m\dot{W}_0 = mU_0 q_0 + Z_0^a + mg = 0$

③ also,

$$I_{yy}(\dot{q}_0 + \dot{q}') = M_0^a + UM_0^a + WM_W^a + \dot{W}M_W^a + qM_q^a$$

and $I_{yy}\dot{q}_0 = M_0^a = 0$

④ also, $I_E(\ddot{\delta}_{E0} + \ddot{\delta}_{E'}) = H_0^E + UH_0^E + WH_W^E + qH_q^E + \dot{W}H_W^E + \delta_E H_{\delta E}$

$$I_E(\ddot{\delta}_{E0} + \ddot{\delta}_{E'}) = \frac{1}{2}\rho V \omega^2 S_E C_E$$

$$[b_0 + b_H a_H + b_E(\delta_{E0} + \delta_{E'}) + b_T \delta_T]$$

and $I_E \ddot{\delta}_{E0} = \frac{1}{2}\rho V \omega^2 S_E C_E (b_0 + b_H a_H + b_E \delta_{E0} + b_T \delta_T)$

⑤ In trim flight, $\dot{\theta} = q$

$\therefore \dot{\theta}_0 + \dot{\theta}' = q + q'$ and $\theta_0 = q_0 = 0$

⑥ at the end, $\dot{\delta}_E = \dot{\delta}_E$

$$\therefore \dot{\delta}_{E0} + \dot{\delta}_{E'} = \dot{\delta}_{E0} + \dot{\delta}_{E'} \quad \text{and} \quad \dot{\delta}_{E0} = 0$$

Other way (more common way to do it)

$$x = [U \ W \ q \ \theta \ \delta_E \ \ddot{\delta}_E]$$

$$M\ddot{x} = N(x) + F(\dot{x}, x) + G(x)$$

For small perturbation theory,

$$\text{at equilibrium point, } M\ddot{x}_0 = N(x_0) + F(0, x_0) + G(x_0) = 0$$

apply a small perturbation,

$$M(\dot{x}_0 + x') = N(x_0 + x') + F(\dot{x}', x_0 + x') + G(x_0 + x') = 0$$

$$\therefore Mx' = x'N_{x_0} + \dot{x}'F_{\dot{x}_0} + x'F_x + x'G_x$$

$$\therefore Mx' = \dot{x}'F_{\dot{x}_0} + x'R_{x_0}$$

Therefore, calculate F_{x_0} and $R_{x_0} \dots$

9. Derive expressions for the hinge moment derivatives (H_U , H_W , H_q , $H_{\dot{W}}$, H_{δ_E}) and the elevator control derivatives (X_{δ_E} , Z_{δ_E} , M_{δ_E}). Use the data provided to evaluate these derivatives at the equilibrium condition specified in table 2.

$$H_E = I_E \ddot{\delta}_E = \frac{1}{2} \rho V_\infty^2 S_E C_E (b_0 + b_H \alpha_H + b_E \delta_E + b_T \delta_T)$$

$$\therefore H_U = \frac{dH_E}{dV_\infty} \frac{dV_\infty}{dU} = \rho U_e S_E C_E (b_0 + b_H \alpha_H + b_E \delta_E + b_T \delta_T)$$

$$H_W = \frac{dH_E}{d\alpha} \frac{d\alpha}{dW} = \frac{dH_E}{d\alpha_H} \frac{d\alpha_H}{d\alpha} \frac{d\alpha}{dW} = \rho U_e S_E C_E b_H \left(1 - \frac{d\epsilon}{d\alpha}\right)$$

$$H_q = \frac{dH_E}{dU_H} \frac{dU_H}{dq} + \frac{dH_E}{dW_H} \frac{dW_H}{dq} = -(Z_H - Z_{CG}) H_{UH} + (X_H - X_{CG}) H_{WH}$$

$$\text{where } H_{UH} = H_U, H_{WH} = \rho U_e S_E C_E b_H$$

$$H_{\delta_E} = \frac{1}{2} \rho V_\infty^2 S_E C_E b_E$$

$$H_{\dot{W}} = \frac{dH_E}{dW_H} \frac{dW_H}{d\epsilon_H} \frac{d\epsilon_H}{d\dot{W}} \quad \text{where} \quad \frac{dW_H}{d\epsilon_H} = -U_e$$

$$\epsilon_H = \epsilon_0 + \frac{d\epsilon}{d\alpha} \left(\alpha - \frac{d\alpha}{dt} \frac{l_H}{U_e} \right) \quad \text{where} \quad \frac{d\alpha}{dt} = \frac{d\alpha}{dW} \frac{dW}{dt} = \frac{1}{U_e} \dot{W}$$

$$= \epsilon_0 + \frac{d\epsilon}{d\alpha} \left(\alpha - \frac{l_H}{U_e^2} \dot{W} \right)$$

$$\therefore \frac{d\epsilon_H}{d\dot{W}} = - \frac{d\epsilon}{d\alpha} \frac{l_H}{U_e^2}$$